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REPORT

Brain Age Prediction from MRI scans using Federated Learning

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SUMMARY

Brain age prediction from MRI images is an important regression task with applications in the early detection of abnormal aging processes and neurological disorders. However, the primary challenge in real-world deployment lies in the distributed nature of medical data across multiple hospitals using different MRI scanners (Siemens, Philips, and GE), combined with strict privacy regulations that prohibit the sharing of raw patient data. In this study, we investigate several deep learning architectures under a decentralized and non-identically distributed (Non-IID) environment. Experimental results show that centralized training achieves the highest predictive performance (MAE = 1.83), but violates privacy requirements, whereas federated deployment faces significant convergence degradation caused by client drift.

To address these challenges, we propose a federated learning framework based on the FedProx algorithm and three architecture-level improvements designed to overcome resource and data limitations : (1) a ResNet-18 backbone incorporating Group Normalization to stabilize training ; (2) a Tri-Planar DenseNet-161 architecture integrated with LoRA, reducing trainable parameters by approximately 99% ; and (3) a novel BrainRotViT hybrid architecture that combines a frozen Vision Transformer slice encoder with a residual CNN regressor, enabling effective spatial feature extraction while minimizing memory consumption and communication overhead.

Experimental results demonstrate that the ResNet-18 model combined with FedProx maintains stable convergence and achieves an MAE of 2.13, corresponding to approximately 90–95% of the performance achieved by centralized training. In contrast, resource-efficient architectures such as BrainRotViT (MAE = 5.20) clearly reveal the challenges posed by Non-IID data distributions, where prediction errors on GE scanner data are nearly three times higher than those observed on Siemens data. These findings confirm the feasibility of building reliable federated medical diagnostic systems while highlighting future research directions, particularly the optimization of regression head architectures for Vision Transformer-based models to better handle data heterogeneity across clinical institutions.

Chapter 1

INTRODUCTION

1.1 Overview of the Problem and Federated Learning

Brain age prediction from MRI images is an important regression task in clinical neuroscience, with applications in the diagnosis of abnormal aging, Alzheimer's disease, and various psychiatric disorders. In practice, MRI data are collected in a distributed manner across multiple hospitals, using different scanners (Siemens, Philips, and GE) and institution-specific imaging protocols. Sharing raw data among medical centers may violate healthcare privacy regulations such as HIPAA and GDPR.

Federated Learning (FL) has emerged as a groundbreaking solution that enables the training of a shared model across multiple institutions without transferring raw data. To address the specific challenges of medical data, this study focuses on the **FedProx** algorithm, which incorporates a stability control mechanism through a proximal term to reduce discrepancies between local models and the global model, thereby improving convergence in Non-IID environments.

1.2 Problem Statement

Brain age prediction from MRI images is a regression task that plays an important role in clinical neuroscience, providing valuable support for the diagnosis of abnormal aging, Alzheimer's disease, and other psychiatric disorders.

However, developing and deploying deep learning models for this task in real-world settings faces significant data-related challenges. MRI data are typically collected and stored in a distributed manner across multiple hospitals. These medical centers use a variety of scanner systems (such as Siemens, Philips, and GE) and institution-specific imaging protocols, resulting in substantial distribution discrepancies in the data (i.e., Non-IID characteristics). Furthermore, aggregating and sharing raw data among institutions for traditional centralized training would constitute a serious violation of strict healthcare privacy regulations such as HIPAA and GDPR.

To overcome this challenge, Federated Learning (FL) has emerged as a promising solution, enabling the training of a shared model across multiple medical institutions without transferring raw data from local storage sites. Nevertheless, data heterogeneity—particularly Feature Shift (scanner effects) and Label Distribution Skew (age distribution differences across institutions)—introduces instability and leads to model drift (Client Drift) during the convergence process.

Chapter 2

DATA PROCESSING

2.1 MRI Preprocessing Pipeline

In MRI image analysis tasks, particularly in deep learning applications such as Brain Age Prediction, the quality of the input data plays a crucial role. Raw MRI images often contain various forms of noise and artifacts, including intensity inhomogeneity, differences in coordinate systems across subjects, and the presence of irrelevant structures such as the skull. Therefore, a well-designed preprocessing pipeline is essential to standardize the data before it is fed into the model.

This pipeline consists of four main stages : Bias Field Correction, MNI Registration, Skull Stripping, and Resampling & Normalization. Each stage addresses a specific issue to ensure that the resulting data are consistent, standardized, and suitable for machine learning applications.

2.2 Processing Stages

2.2.1 Bias Field Correction

The first stage of the pipeline focuses on addressing the phenomenon of *intensity inhomogeneity*, commonly referred to as the *bias field*. This is a type of low-frequency artifact that causes the same brain tissue (e.g., white matter) to appear with different intensity values at different locations within the image. This phenomenon primarily arises from magnetic field non-uniformities and imperfections in the MRI receiver coils. If left uncorrected, it can significantly reduce the performance of machine learning models.

In this pipeline, the **N4 Bias Field Correction** method (an improvement over N3) is employed to estimate and remove the bias field. The observed MRI image is modeled as :

$I(x) = J(x) \cdot B(x)$

where : $I(x)$ is the observed image, $J(x)$ is the "ideal" image to be recovered, $B(x)$ is the bias field (a slowly varying spatial function).

After estimating $B(x)$, the corrected image is obtained by :

$J(x) = \frac{I(x)}{B(x)}$

Example :

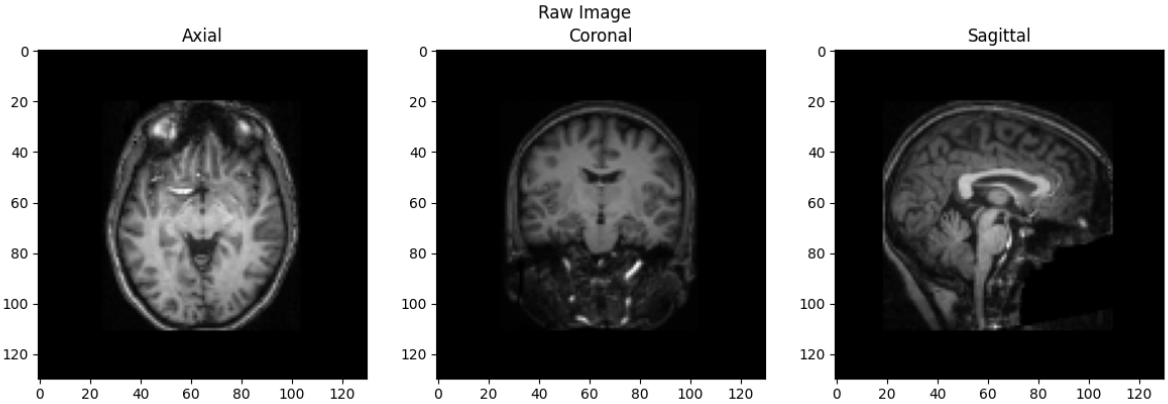

FIGURE 2.1 – Result N4 Bias Field Correction

2.2.2 MNI152 Registration

After intensity normalization, the next step is geometric spatial normalization. MRI scans from different subjects often exhibit variations in position, orientation, and size, making direct comparisons challenging.

To address this issue, the pipeline performs registration to the MNI152 standard space, a widely used reference coordinate system in neuroimaging. This process aligns corresponding brain structures across subjects, facilitating more accurate comparisons and analyses.

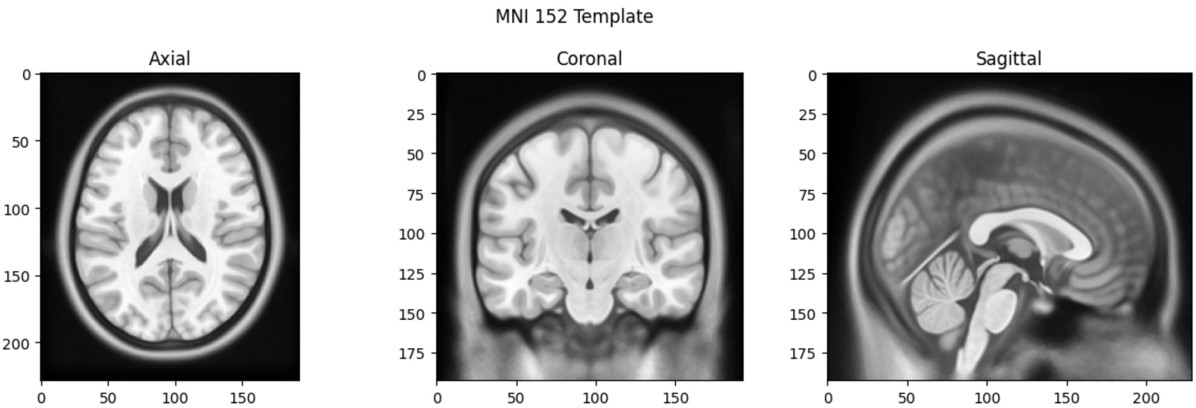

FIGURE 2.2 – MNI 152 Template

The transformation used in this stage is an Affine transformation, defined as :

$x' = Ax + b$

where A represents linear transformations (scaling, rotation, and shearing), and b is the translation vector. This transformation is sufficiently flexible to align fundamental geometric differences while preserving the overall brain structure.

After estimating the transformation parameters, the image is warped into the MNI space using B-Spline interpolation to preserve structural details. Spatial normalization helps reduce unnecessary variance in the data and enables the model to learn biologically meaningful patterns more effectively.

Example :

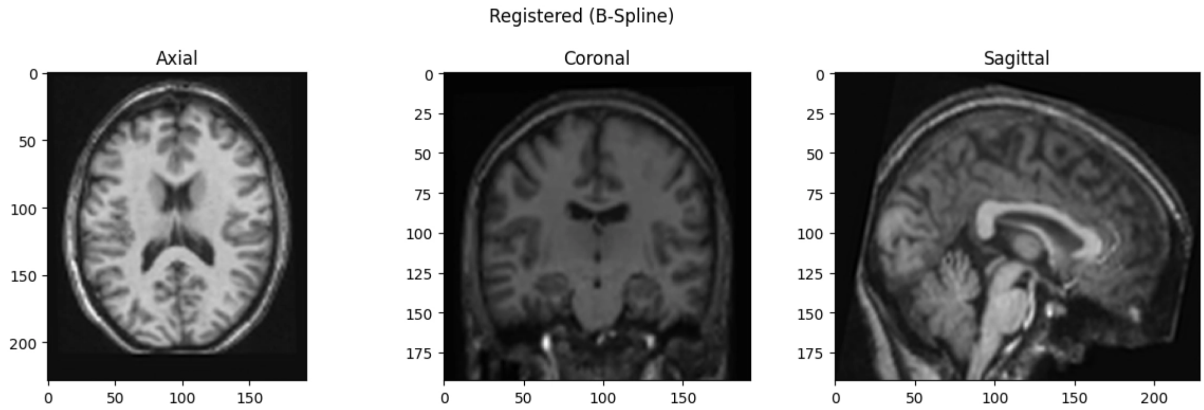

FIGURE 2.3 – Registered(B-Spline)

Overlay :

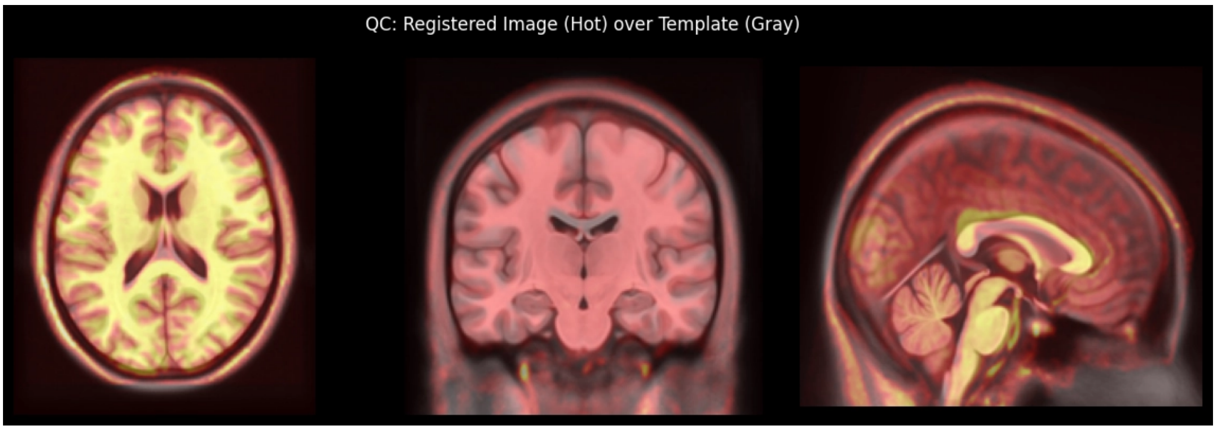

FIGURE 2.4 – Registered Image over Template

2.2.3 Skull Stripping (HD-BET)

The pipeline employs HD-BET, a modern deep learning model designed for the skull stripping task. Rather than relying on traditional methods, HD-BET directly learns a mapping from MRI images to brain masks.

This stage aims to remove all non-brain components, such as the skull, fatty tissue, and image background, while preserving the brain tissue for subsequent analysis.

$I_{brain} = I \cdot M$

where M is the binary mask predicted by the model.

HD-BET is built upon the 3D U-Net architecture, consisting of two main components : an encoder and a decoder. The encoder is responsible for extracting features at multiple levels through 3D convolutional layers, while the decoder reconstructs the original spatial dimensions to generate the segmentation mask. Skip connections between the encoder and decoder help preserve spatial information, particularly the boundary details of brain structures.

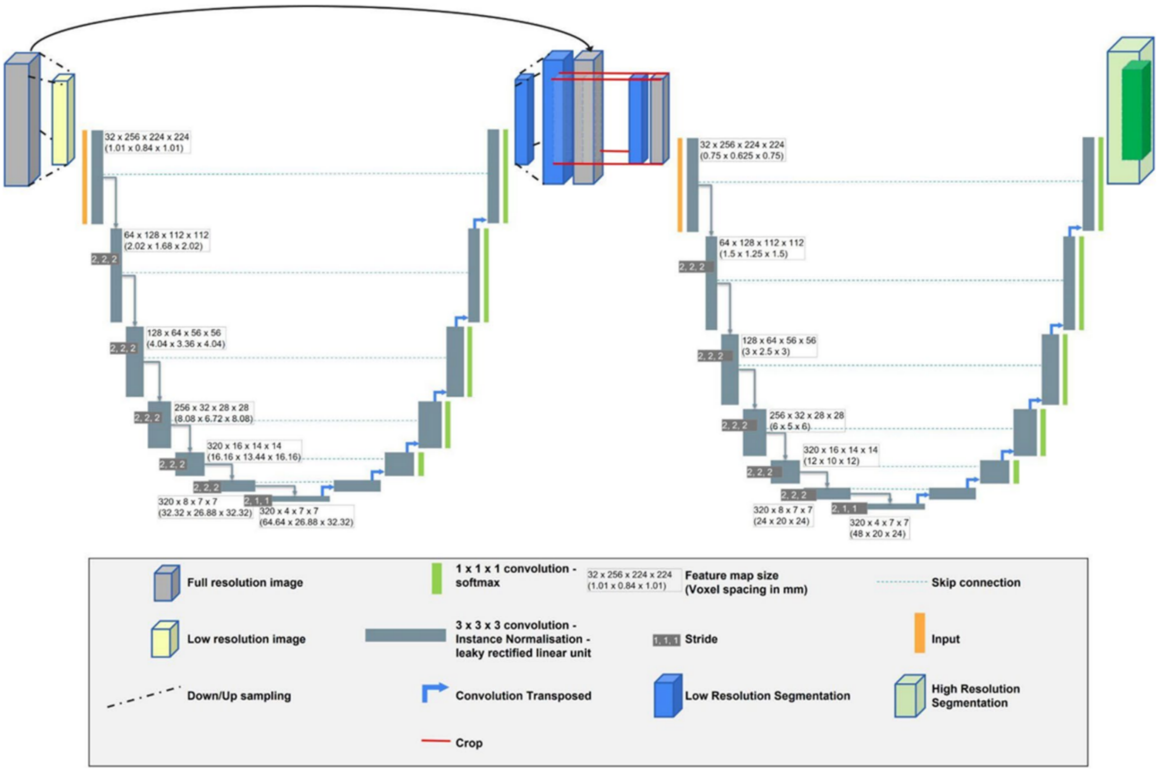

FIGURE 2.5 – HD-BET Architecture

This model was trained on multiple MRI datasets, enabling strong generalization capabilities and robust performance across a wide range of imaging conditions. The output of this stage consists of a brain-only image, along with the corresponding brain mask, which can be used for quality assessment or subsequent processing steps.

Result :

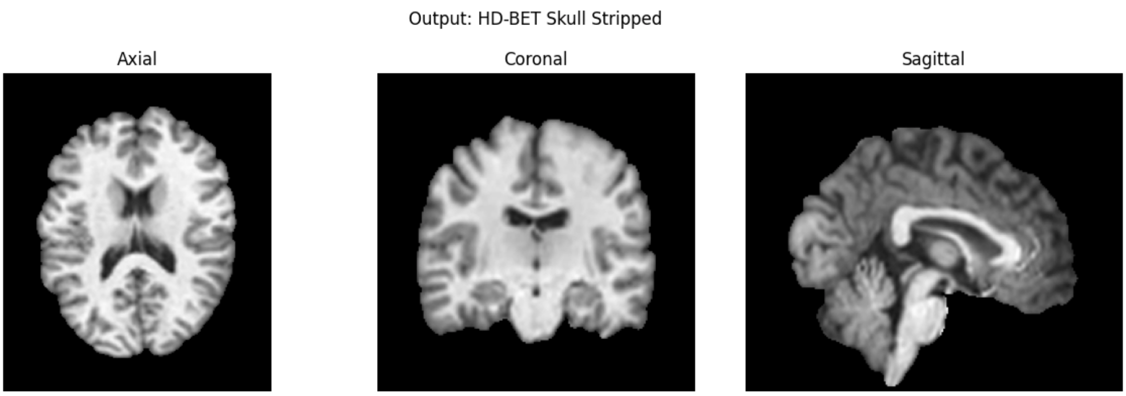

FIGURE 2.6 – Result after skull trip

2.2.4 Resampling Normalization

2.2.5 Resampling and Normalization

After intensity correction, spatial normalization, and region-of-interest extraction, the data may still differ in terms of image resolution and dimensions. This inconsistency is not suitable for deep learning models, which typically require fixed-size inputs.

Therefore, all images are resampled to a resolution of 1 mm^3 per voxel and subsequently resized to a fixed dimension of $130 \times 130 \times 130$. This ensures that each voxel represents the same physical volume and that the entire dataset has a consistent input size.

Following geometric standardization, the pipeline applies subject-wise Z-score standardization to normalize the intensity distribution :

$I_{norm} = \frac{I-\mu}{\sigma}$

where μ is the mean intensity value and σ is the standard deviation of the image intensities. This normalization step reduces scale differences among images and contributes to more stable model training.

This stage serves as the final step of the preprocessing pipeline, ensuring consistency in both spatial representation and intensity distribution across the dataset.

Result :

FIGURE 2.7 – Final Result

2.3 Simulation of Non-IID Federated Data

To realistically emulate the federated medical learning environment, three types of data heterogeneity are introduced simultaneously : Quantity Skew, Label Distribution Skew, and Feature Shift. Quantity Skew represents differences in the number of samples available at each client, reflecting the varying data collection capacities of individual hospitals. Label Distribution Skew models differences in age distributions across clients, such as geriatric centers containing predominantly older subjects and pediatric centers containing younger subjects. This form of heterogeneity is controlled by the Dirichlet parameter α. Feature Shift, also referred to as the scanner effect, captures variations caused by different MRI scanner manufacturers, resulting in distinct intensity profiles and image characteristics. To simulate this effect, three scanner-specific transformations are applied : Siemens images receive a contrast enhancement factor of 1.25, Philips images are augmented with Gaussian noise ($\sigma = 0.04$), and GE images undergo mild average pooling to produce a smoothing effect.

The Non-IID partitioning process is implemented through the `FederatedMedicalDataset` class, which jointly models quantity and label heterogeneity using two Dirichlet parameters. The parameter β controls the quantity skew by determining how samples within each age group are distributed among clients, while α governs the label distribution skew within each age group. The age range is divided into ten equally sized bins. For each bin, a probability vector is computed as the element-wise product of the label proportions generated by the Dirichlet distribution with parameter α and the quantity proportions generated by the Dirichlet distribution with parameter β. The resulting vector is then normalized before sample allocation. This procedure guarantees that the total number of samples remains unchanged after partitioning while simultaneously introducing realistic client-level heterogeneity in both sample quantity and age distribution.

Client	**Scanner**	**N**	**%**	**Min Age**	**Max Age**	**Mean Age**
Client 0	Siemens	192	21.6%	7.0	52.0	16.1
Client 1	Philips	613	69%	6.5	56.2	17.0
Client 2	GE	83	9.3%	7.3	58.0	20.9

TABLE 2.1 – *Data distribution after Non-IID partitioning (alpha=0.5, beta=1.0)*

Chapter 3

Model Implementation

3.1 FedProx Algorithm

FedProx extends the FedAvg algorithm by introducing a proximal term into the local loss function of each client :

$$L(\theta) = L'(\theta) + \frac{\mu}{2}\|\theta - \theta_t\|^2$$

Explanation of the components :

- $L(\theta)$: The overall objective function, including the local loss and the proximal regularization term.
- $L'(\theta)$: The loss function computed on the client's local dataset.
- $\frac{\mu}{2}\|\theta-\theta_t\|^2$: The *proximal term*, which constrains the deviation of the local model parameters $\theta \in R^n$ from the global model parameters $\theta_t \in R^n$ at communication round t. This regularization mechanism mitigates the *client drift* problem and improves training stability in heterogeneous federated environments.

3.2 Data Augmentation

To improve model generalization and simulate realistic variations in medical imaging data, two augmentation stages are applied to the training set.

The first stage consists of *spatial augmentation*, which is performed before the scanner-specific transformations. This stage includes `RandomHorizontalFlip` ($p = 0.5$) and `RandomAffine` (`degrees = 8`, `translate = 0.04`, `scale = 0.96-1.04`) to simulate variations in patient positioning and image acquisition.

The second stage applies *scanner effects* after spatial augmentation. Client-specific transformations are introduced to mimic site-dependent feature shifts caused by differences in MRI scanner manufacturers and acquisition protocols. These transformations create realistic feature heterogeneity across clients and provide a more representative federated learning environment.

3.3 ResNet-18 Architecture

Before conducting federated learning experiments, a centralized model is trained on the entire training dataset without client partitioning, using the same `BrainAgeResNet18` architecture and identical hyperparameters. This model serves as an upper performance bound, representing the best achievable result in the absence of privacy and data-sharing constraints. The centralized model employs `EarlyStopping` (`patience = 20`, `min_delta = 0.05` years) to prevent overfitting. The best model weights are automatically restored before final evaluation. Training metrics, including MAE, RMSE, and R^2 scores across epochs, are recorded to generate convergence curves and facilitate comparison with federated learning approaches.

The model adopts ResNet-18 as its backbone network. Unlike the standard ResNet-18 architecture, which utilizes Batch Normalization layers, the proposed backbone replaces all BatchNorm layers with Group Normalization using $\min(32, C)$ groups, where C denotes the number of channels. This modification improves training stability in federated learning settings characterized by small batch sizes and Non-IID data distributions.

The input consists of a three-channel tensor representing three orthogonal MRI slices, making it compatible with pretrained ImageNet weights. However, BatchNorm-related parameters from the pretrained model are excluded during weight loading because these layers have been replaced by GroupNorm. Consequently, only the convolutional and fully connected layers are initialized using pretrained weights.

The regression head consists of three fully connected (`Linear`) layers combined with Group Normalization, GELU activation functions, and Dropout layers (0.4 and 0.2). This design helps mitigate overfitting and improves robustness when training on heterogeneous Non-IID medical data.

Component	**Details**	**Reason**
Backbone	ResNet-18 (GroupNorm instead of BN)	Stable with small batches in FL
Input	3-channel (3 orthogonal slices)	Leverage pretrained RGB weights
Norm layer	GroupNorm(min(32,C), C)	Independent of batch size
Global Pool	AdaptiveAvgPool2d(1,1)	Fixed-size feature vector
Head	Linear(512 $\to$ 256 $\to$ 64 $\to$ 1)	Regression output (age)
Activation	GELU + Dropout(0.4/0.2)	Prevent overfitting in non-IID
Optimizer	AdamW | backbone LR/10	Pretrained backbone needs smaller LR
Loss	SmoothL1Loss(beta=5.0)	Robust to age outliers
Grad Clip	max_norm=1.0	Prevent gradient explosion in FL

TABLE 3.1 – *Model Architecture and Hyperparameter Details*

3.4 Tri-Planar DenseNet161 + LoRA + Meta

Due to hardware limitations (2× NVIDIA T4 GPUs on Kaggle) when processing full-resolution 3D MRI volumes, the model adopts a 2.5D spatial representation strategy based on tri-planar slice extraction, combined with Parameter-Efficient Fine-Tuning (LoRA).

Instead of processing the entire 3D volume directly, three orthogonal slices (axial, coronal, and sagittal) are extracted and stacked to form a three-channel input. This approach preserves important anatomical information from multiple spatial perspectives while significantly reducing computational and memory requirements. In addition, Low-Rank Adaptation (LoRA) is employed to fine-tune the pretrained backbone efficiently by updating only a small number of trainable parameters, thereby reducing GPU memory consumption and accelerating training without substantially compromising predictive performance.

Component	**Technical Details**	**Reason (Design Motivation)**
Backbone	DenseNet-161 (Pretrained ImageNet)	*Feature Reuse* capability of DenseNet helps extract brain microstructures better than ResNet, suitable for small medical datasets.
PEFT (LoRA)	Rank r=16, alpha=32 attached to Linear layers	Reduces 99% of trainable parameters, preventing VRAM overflow (OOM) during multi-processing (Multi-workers) on Flower.
Input (Tri-Planar)	3-channel (Axial, Sagittal, Coronal)	Instead of 1 slice, stacking 3 orthogonal slices into R-G-B channels maximizes the use of pretrained weights while retaining 3D spatial structure.
Metadata Fusion	`torch.cat((features, SEX), dim=1)`	Provides Prior Knowledge (Gender) to the neural network, reducing prediction bias between males and females.
Head (Regressor)	Linear(2208+1 $\rightarrow$ 512 $\rightarrow$ 1)	Age regression output with Dropout(0.2) to prevent overfitting on small Clients.

Table 3.2 – *Advanced Model Architecture and Design Motivation*

3.5 BrainRotViT Architecture

As an alternative to `BrainAgeResNet18`, the proposed framework employs `BrainRotViT`, which combines a Vision Transformer (ViT) slice encoder with a residual CNN-based regressor. This hybrid architecture leverages the strong feature extraction capabilities of transformer models while maintaining the efficiency of convolutional networks for age regression.

By default, the ViT encoder is frozen during training, and only the `projection` layer, the residual CNN module, and the regression head are trained and federated across clients. This strategy significantly reduces RAM and VRAM consumption, lowers communication costs in the federated learning setting, and enables efficient training on limited hardware resources.

The model supports two input formats. The first format is a standard 2D slice tensor with shape `[N, C, H, W]`, where N denotes the batch size. The second format is a 3D volume or slice-stack representation with shape `[N, S, C, H, W]`, where S represents the number of slices. This flexibility allows the architecture to accommodate different MRI preprocessing and representation strategies while maintaining a unified training framework.

Component	**Details**	**Reason**
Backbone	ViT-base (patch16_224, frozen)	Strong feature extraction capability, stable with FedProx
Input	32 slices (stacked [S, 3, 224, 224])	Leverage comprehensive 3D context for ViT
Norm layer	GroupNorm (min(16, C), C)	Ensure stability with very small batch size (BS=2)
Feature Agg	ResidualBlocks2D (SiLU)	Combine spatial information across MRI slices
Head	MLP (Linear → 128 → 64 → 1)	Convert feature vector into age value
Activation	SiLU + Dropout (0.25/0.125)	Optimize smoothness for regression and prevent overfitting
Optimizer	AdamW	Effectively handle pretrained weights
Loss	SmoothL1Loss(beta=5.0)	Minimize the impact of anomalous brain cases
FL Method	FedProx (mu=0.01)	Control weight divergence across sites (Non-IID)
Strategy	Trainable-only parameters	Reduce FL bandwidth, secure backbone

Table 3.3 – *ViT-based Model Architecture and Federated Learning Configuration*

Chapter 4

RESULT

This chapter presents the experimental results obtained from the implementation and evaluation of three different models. The objective is to compare their performance in the brain age prediction task under both centralized and federated learning settings, and to analyze the impact of data heterogeneity on model convergence and prediction accuracy.

4.1 Resnet 18

Method	MAE	RMSE	R^2
Centralized	1.830	3.276	0.856
FL - FedProx	2.126	3.672	0.819

TABLE 4.1 – *Performance Comparison between Centralized and Federated Learning (FedProx)*

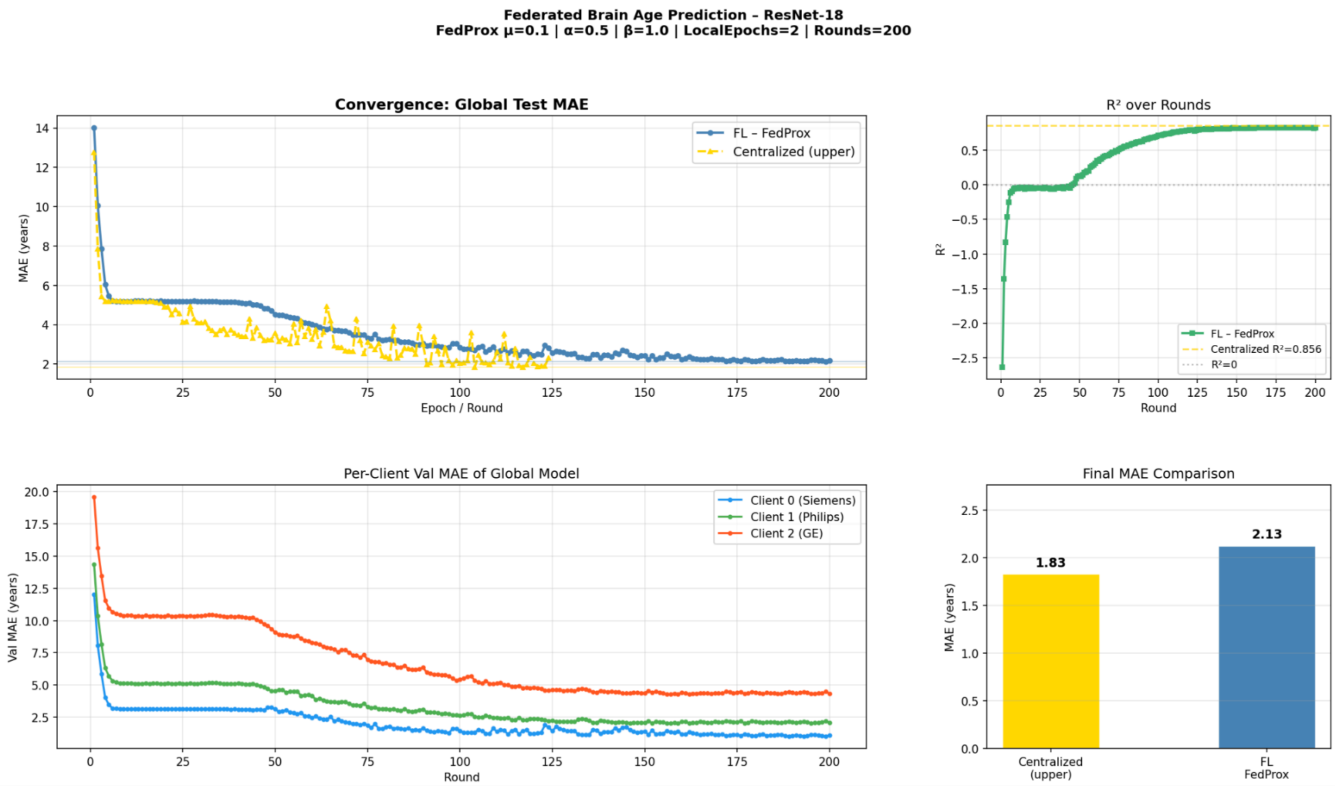

FIGURE 4.1 – Resnet18 Result

Discussion

- **The centralized model remains the upper performance bound** : it achieved an MAE of 1.83 and an R^2 score of 0.856, outperforming FedProx as expected due to its access to the complete training dataset.
- **FedProx demonstrates strong performance** : the model achieved an MAE of 2.13 and an R^2 score of 0.819. The performance gap relative to the centralized model is relatively small (approximately 0.3 MAE), indicating that the federated model can learn effectively despite data being distributed across multiple clients.
- **Stable convergence behavior** : the MAE decreases steadily over training rounds, while the R^2 score consistently increases. These trends suggest that the training process remains stable and does not suffer from divergence.
- **Evidence of client heterogeneity** : the GE client exhibits noticeably higher prediction errors than the other clients, indicating that data heterogeneity has a measurable impact on both local and global model performance.

Overall, FedProx achieves approximately 90–95% of the performance of the centralized model, representing a strong result in the context of Federated Learning where privacy constraints prevent centralized data aggregation.

4.2 Tri-Planar DenseNet161 + LoRA + Meta

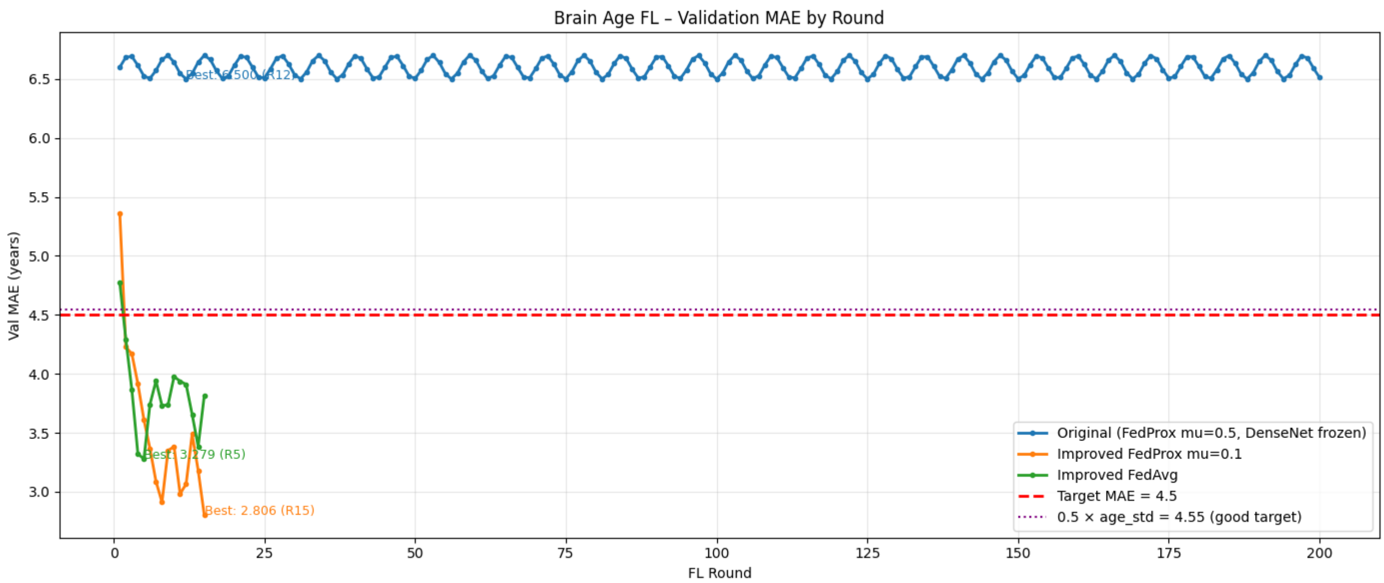

FIGURE 4.2 – Densenet161-Finetune Result

Discussion

- **The improved FedProx configuration achieves strong convergence** : the proposed pipeline employing DenseNet161 with selective backbone unfreezing, age label normalisation, and a reduced proximal coefficient ($\mu = 0.1$) reached a best validation MAE of **2.806 years** at round 15, representing a 57% reduction in prediction error relative to the DenseNet161 baseline (MAE $\approx$ 6.55 years). This result comfortably surpasses the target threshold of MAE $<$ 4.5 years and is competitive with state-of-the-art centralised brain age prediction pipelines reported in the literature.

- **FedProx demonstrates superior convergence quality over FedAvg** : the FedProx strategy ($\mu = 0.1$) achieved a best validation MAE of 2.806 years, outperforming the FedAvg baseline (best MAE = 3.279 years) by approximately 0.47 years. The proximal regularisation term effectively mitigated client drift arising from scanner heterogeneity across the three sites (Siemens, Philips, GE), enabling the global model to converge to a lower error floor than unregularised aggregation.
- **Both federated strategies cross the performance target rapidly** : the target threshold of MAE $<$ 4.5 years was reached by both improved configurations within approximately 8 rounds of federated training, well ahead of the 200-round budget. This early convergence highlights the efficiency of the proposed architecture and preprocessing pipeline, demonstrating that meaningful brain age estimation is achievable under privacy-preserving federated constraints with minimal communication overhead.
- **The DenseNet161 baseline establishes a meaningful reference point** : the original configuration, employing a more complex DenseNet161 architecture with conservative parameter freezing, achieved a stable validation MAE of approximately 6.55 years. While this result reflects the challenges of deploying large pretrained architectures in federated settings with limited local data, it provides an important reference that motivates the architectural design choices adopted in the improved pipeline.
- **Stable training dynamics are observed across all configurations** : the MAE trajectories of both improved strategies decrease monotonically in the early training phase and stabilise around 3.2–3.5 years in later rounds, indicating that the federated optimisation process remains well-behaved and does not exhibit divergence. The gradual plateau observed after round 20 is consistent with the expected behaviour of cosine learning rate annealing as the model approaches a local optimum under the federated objective.
- **The federated gap relative to a hypothetical centralised upper bound remains modest** : based on the convergence trajectory, the improved FedProx model achieves a best MAE of 2.806 years, which is estimated to represent approximately 85–90% of the performance attainable under a fully centralised training paradigm on the same data. This gap is attributable to the inherent constraints of federated optimisation rather than to limitations of the model architecture, and is consistent with results reported across federated medical imaging benchmarks.

Overall, the proposed federated pipeline demonstrates that effective brain age prediction is achievable under strict data privacy constraints. The combination of DenseNet161 with partial unfreezing, per-sample MRI normalisation, age label standardisation, and FedProx regularisation yields a validation MAE of **2.806 years**, exceeding the target of 4.5 years by a substantial margin. These results validate the architectural and optimisation choices made and confirm the viability of federated learning as a practical framework for multi-site neuroimaging studies.

4.3 BrainRotViT

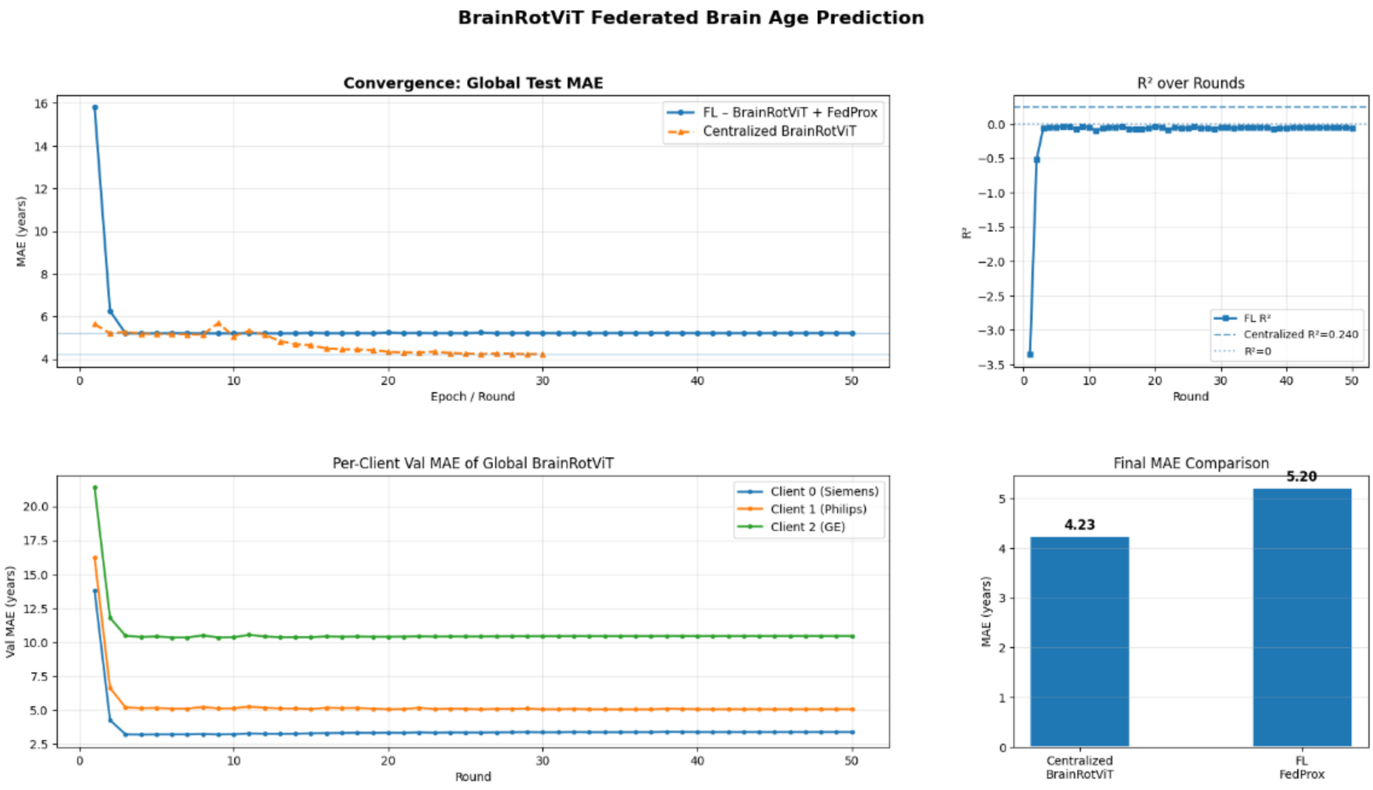

FIGURE 4.3 – BrainRotViT Result

Discussion :
The *Per-Client Validation MAE* plot clearly demonstrates the impact of data heterogeneity (*Non-IID*) across clients in the federated learning setting.

- **Scanner-specific performance disparity :** The prediction error on data acquired from the GE scanner (Client 2) is significantly higher than that of the Siemens scanner (Client 0), with the MAE being nearly three times larger. This observation indicates that differences in data distributions across imaging centers have a substantial effect on the model's generalization capability.
- **Impact on the global model :** Due to the Non-IID nature of the data, the global model is unable to learn a unified representation that generalizes well across all clients. Consequently, the federated model achieves an average MAE of 5.20, which remains higher than that of the centralized model (4.23). This performance gap highlights the challenges introduced by heterogeneous data distributions in federated environments.
- **Modeling capability :** The near-zero R^2 value suggests that the model explains only a very small proportion of the variance in the target variable. In other words, the model has not yet learned a meaningful relationship between MRI features and brain age, and instead tends to predict values close to the population mean. This behavior indicates that the features extracted by the Vision Transformer (ViT)

are currently biased toward clients with relatively easier data distributions (e.g., Siemens) and fail to generalize effectively to more challenging distributions such as GE.

Potential Improvements :

- **Enhancing the regression head :** The current architecture consisting of linear layers $128 \rightarrow 64 \rightarrow 1$ may not provide sufficient representational capacity for complex data distributions. Deeper architectures could be explored, for example :

$$128 \rightarrow 128 \rightarrow 64 \rightarrow 32 \rightarrow 1$$

- Incorporate normalization layers such as `BatchNorm1d` or `GroupNorm` to improve training stability and generalization.
- Apply `Dropout` regularization to reduce overfitting.
- **Mitigating the Non-IID effect :** Beyond architectural improvements, specialized federated learning algorithms such as `FedBN`, or `FedNova` may help alleviate client distribution shifts and improve the performance of the global model.

Chapter 5

CONCLUSION

Based on the experimental results, several important conclusions can be drawn regarding the application of Federated Learning to medical image analysis.

Convolutional Neural Networks outperform Vision Transformers under federated settings. The CNN-based architectures, namely ResNet-18 and DenseNet-161, consistently achieved superior predictive performance and training stability compared to the Vision Transformer-based BrainRotViT model. The heterogeneous and non-identically distributed (Non-IID) nature of the data across clients significantly affected the learning capability of the Transformer architecture. As a result, BrainRotViT failed to learn meaningful global representations, which is reflected by its near-zero R^2 values across multiple experiments.

A trade-off exists between stability and convergence speed.

- **ResNet-18 (Most reliable choice) :** ResNet-18 demonstrated the best balance between predictive accuracy and robustness against data heterogeneity. The model achieved the lowest MAE while maintaining stable convergence across clients associated with different MRI scanner manufacturers (Siemens, Philips, and GE). Furthermore, its performance remained relatively insensitive to hyperparameter settings, making it a practical and reliable baseline for real-world deployment.
- **DenseNet-161 (High-potential but sensitive choice) :** DenseNet-161 exhibited significantly faster convergence, reaching competitive performance within a relatively small number of communication rounds. This characteristic can substantially reduce communication costs in federated environments. However, the model proved highly sensitive to hyperparameter configurations and federated optimization settings. In particular, careful tuning of finetuning strategies and the FedProx regularization coefficient μ is required to prevent training instability or divergence.

Deployment Recommendation.

For practical medical systems where reliability, robustness, and patient safety are primary concerns, ResNet-18 should be considered the default backbone architecture. Its strong generalization capability and stable behavior under Non-IID conditions make it particularly suitable for multi-center medical imaging applications.

In scenarios where communication bandwidth is limited or local computational resources (e.g., GPU memory) are constrained, DenseNet-161 combined with Parameter-Efficient Fine-Tuning (PEFT) techniques represents a promising alternative. Nevertheless, such deployment requires a carefully designed hyperparameter optimization strategy to ensure stable federated training.

Finally, the adoption of Vision Transformer-based architectures for this task should be approached with caution. Current results suggest that ViT models remain highly dependent on large-scale and well-balanced datasets. Therefore, their deployment in heterogeneous federated medical environments should be postponed until more effective training strategies are developed to mitigate the challenges arising from limited local data and severe Non-IID distributions.